Predicting the efficacy of malaria vaccines across the life-course

Supplementary Notebook 3. Modelling seasonality in transmission rate

Nicholas J. Savill

Philip J. Spence

We use the circular von Mises distribution to model seasonal changes in transmission rate. The Von Mises distribution for angle θ is given by

$$f(\theta \mid \mu, \kappa) \propto \exp(\kappa \cos(\theta - \mu))$$

where μ is the angle at the peak of the distribution. We place the peak of the season at week S_{peak} relative to last primary vaccination. κ is a measure of the concentration of probability around μ. $\kappa = 0$ corresponds to a uniform circular distribution. Let $S_{\text{width}} = e^{-\kappa} \in [0, 1]$. $S_{\text{width}} = 1$ corresponds to no seasonality (i.e., constant transmission rate through time), and as S_{width} approaches zero the malaria season becomes shorter.

Therefore, the blood infection rate, λ, in week t post last primary vaccination is modified by the factor

$$\frac{\exp\left(-\ln(S_{\text{width}}) \cos\left(2\pi \left[\frac{t - S_{\text{peak}} + 26}{52}\right]\right)\right)}{Z}$$

where $Z = \sum_{i=0}^{51} \exp\left(-\ln(S_{\text{width}}) \cos\left(2\pi \left[\frac{t - S_{\text{peak}} + 26}{52}\right]\right)\right)$ is a normalisation factor.

Roca-Feltrer et al. (2009) defined "marked seasonality" if 75% or more of all episodes occurred in six or less months of the year. We define the length of the malaria season as the minimum number of months that contain 95% of the total malaria cases.

This definition only holds when there is a well de-marked malaria season.

The blue line in Figure 1 shows the transmission rate modification factor for a value of $S_{\text{width}} = 0.11$, which was estimated from the phase 3 RTS,S and SMC trial in Dicko et al. (2024) (Notebook S8). This results in a season length of 6.3 months according to our definition. The malaria season is represented by the grey region which contains 95% of cases.

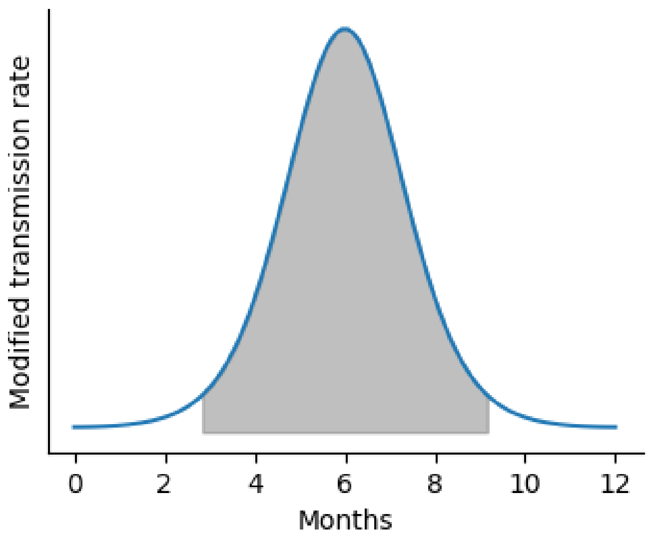

Figure 1: The modified transmission rate (blue line). 95% of infections occur within the grey area.

Figure 2 shows how season length (as defined above) changes for different values of S_{width}.

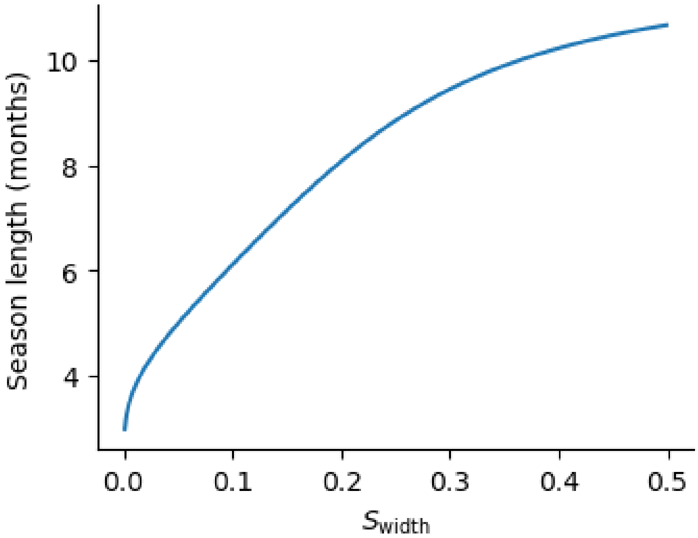

Figure 2: A line plot

Bibliography

Dicko, Alassane, Jean Bosco Ouedraogo, Issaka Zongo, Issaka Sagara, Matthew Cairns, Rakiswendé Serge Yerbanga, Djibrilla Issiaka, et al. 2024. "Seasonal Vaccination with RTS,S/AS01E Vaccine with or Without Seasonal Malaria Chemoprevention in Children up to the Age of 5 Years in Burkina Faso and Mali: A Double-Blind, Randomised, Controlled, Phase 3 Trial". *Lancet Infectious Diseases* 24 (1): 75–86. https://doi.org/10.1016/S1473-3099(23)00368-7.

Roca-Feltrer, Arantxa, Joanna RM Armstrong Schellenberg, Lucy Smith, and Ilona Carneiro. 2009. "A Simple Method for Defining Malaria Seasonality". *Malaria Journal* 8 (December):276–90. https://doi.org/10.1186/1475-2875-8-276.